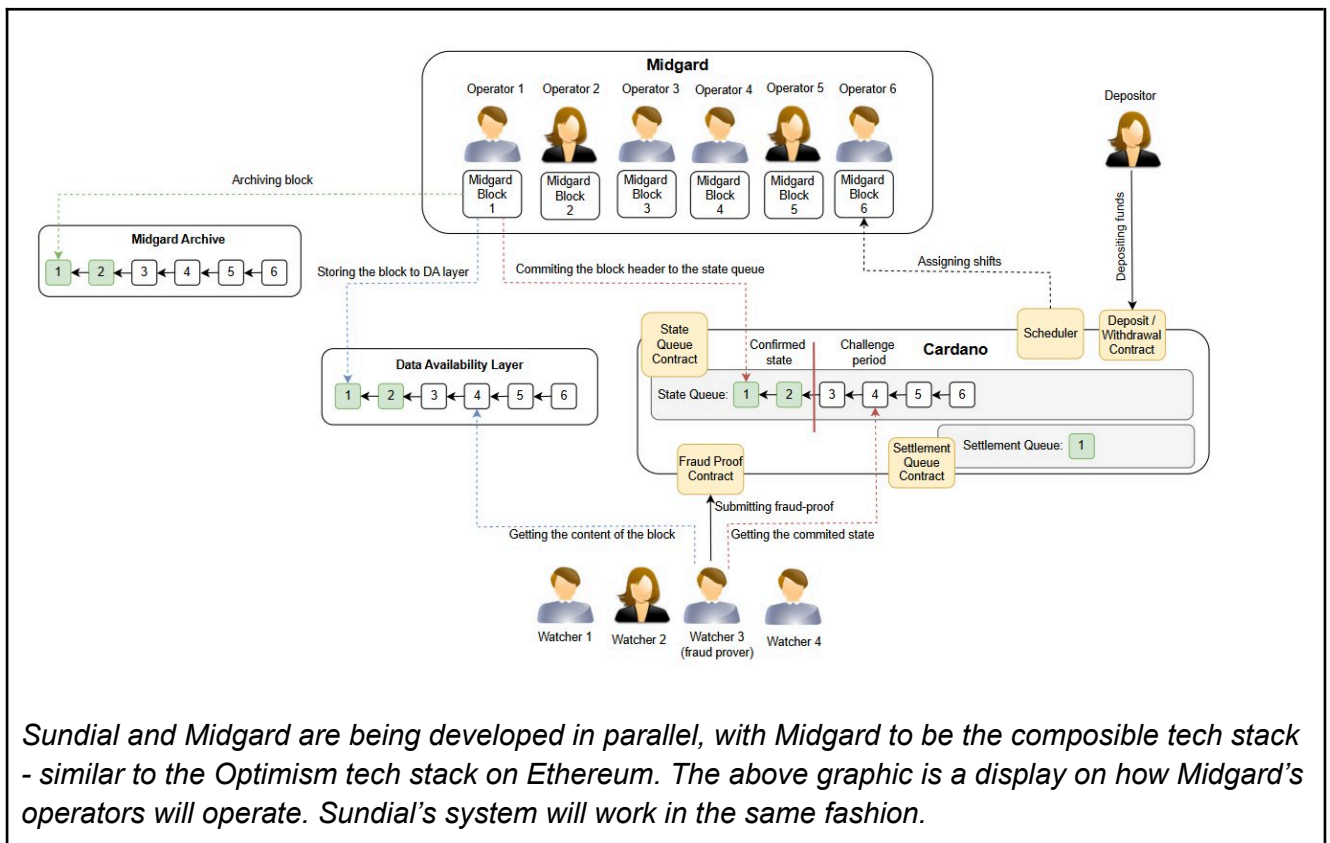


Sundial Litepaper

Working Draft - July 21st, 2025

Sundial is the first optimistic rollup network on Cardano - custom built to serve as Bitcoin's utility and yield layer. It offloads transaction processing from Layer 1 (L1) to Layer 2 (L2) while maintaining L1's security and decentralization properties for those transactions. As a result, it can handle a significantly higher volume of transactions without compromising on security. This whitepaper outlines the interactions between L1 and L2, as well as going into detail about Sundial's specific target market fit: being a hybrid L2 for other UTxO chains, including Bitcoin, and scaling new and novel use cases as a "super" UTxO chain.

The optimistic rollup model implemented in Sundial represents a significant advancement in the scalability and security of general-purpose Layer 2 solutions on Cardano. Below are detailed economic incentives that ensure the protocol's integrity and efficiency.



Operators guarantee the validity of their committed blocks by posting bonds, which are slashable if the blocks are proven invalid. Members of the public are incentivized to watch the operators' blocks, and they are rewarded with a portion of the slashed bonds when their fraud proofs prevent invalid blocks from merging into Sundial's confirmed state. Sundial's architecture leverages a family of smart contracts on Cardano's L1 to manage state transitions and enforce the security of the L2 operations. Key components include a robust operator management system, efficient storage for transaction blocks committed from the L2, and a comprehensive mechanism for submitting and validating fraud proofs. Overall, Sundial's design aims to provide a scalable and secure solution for the growing needs of the Bitcoin and Cardano ecosystems and beyond.

Technical Overview:

Sundial is a Layer 2 (L2) scaling solution for the Bitcoin and Cardano blockchains. It employs optimistic rollup technology to enhance Cardano's base capacity to process transactions and host more complex applications, delivering a richer user experience at a more competitive cost. As Cardano continues to grow in usage and demand, scaling solutions like Sundial are critical for maintaining high performance and low transaction costs.

Optimistic rollups process blocks of transactions off-chain and commit those blocks' headers onchain to the L1 ledger. Each block is committed by a Sundial operator who guarantees the block's validity. The block then waits in a queue for at least a fixed duration to be merged into the confirmed state of the optimistic rollup on L1. The operator must collateralize their guarantee with a bond deposit and publish the full contents of the block on the publicly accessible data availability (DA) layer. While a committed block is queued, anyone can inspect its contents on the data availability layer and ascertain whether it is valid. If someone detects that the block is invalid, they can submit a fraud proof to prevent it from being merged into the confirmed state, slash the operator's bond, and receive a portion of the forfeited bond as a reward.

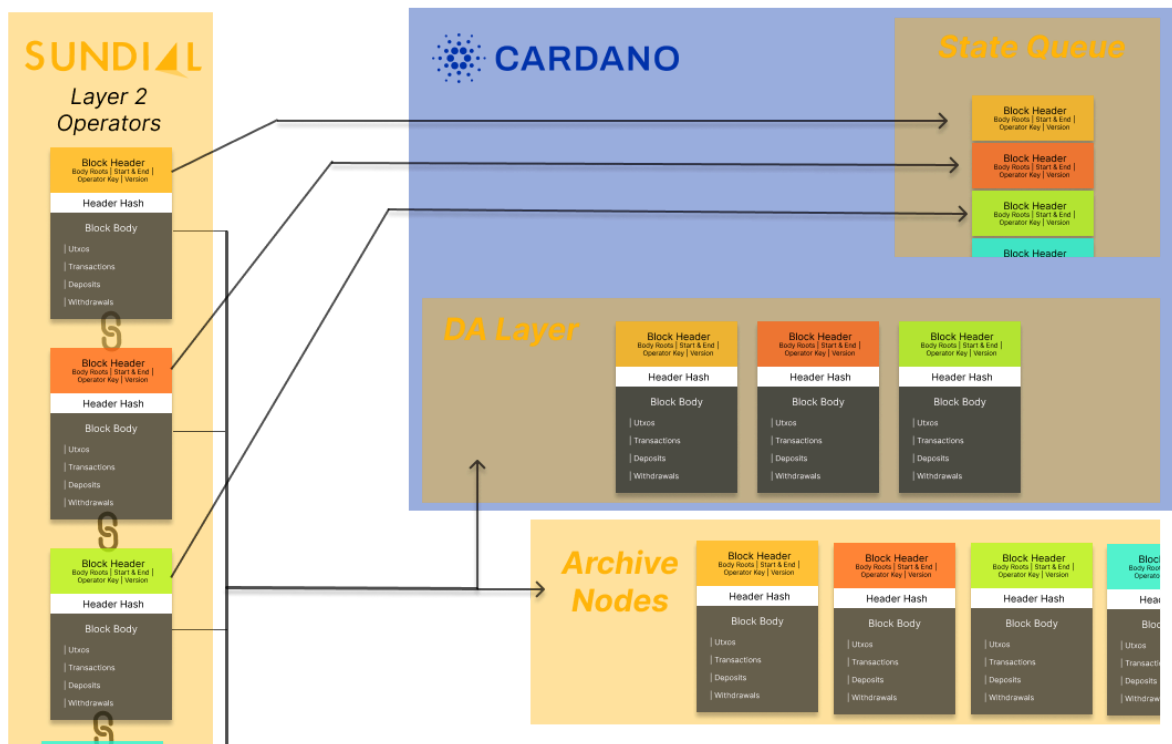
An optimistic rollup can process a large number of transactions offchain while maintaining 1-of-n security and finality properties that are similar to transactions processed directly onchain, as long as:

- The bond requirement for the rollup's operators is large enough to discourage fraud.
- The reward for preventing an invalid block from merging is large enough to encourage public vigilance in watching the operators.
- The waiting period for committed blocks is long enough to allow the watchers to detect and prove fraud before those blocks are merged.
- The data availability layer is accessible by anyone who wishes to inspect the rollup blocks at any time that they wish to do so.

Whenever the latter three security parameter values are calibrated to provide a high probability of invalid blocks being detected and disqualified, the bond requirement is a strong deterrent against operators attempting fraud. An operator cannot dismiss the forfeited bond as merely a "cost of doing business" paid to obtain potentially larger revenues from fraud. Whenever an invalid block is disqualified, it does not affect the confirmed state, so there are no revenues from that fraud to offset the operator's forfeited bond.

The main design goal of Sundial is to streamline the processes by which blocks are committed/merged, fraud is detected, and fraud proofs are verified onchain. Advancing this goal allows the security parameters to be calibrated to achieve a better balance between security, transaction throughput, confirmation time, and community participation in committing blocks and detecting fraud.

Scalability and Efficiency



By processing transactions off-chain and only validating them on-chain when fraud proofs challenge them, Sundial significantly increases throughput and reduces costs for Cardano transactions. Its rolled up block headers use sparse Merkle trees and compact state representations to enhance the protocol's efficiency further, enabling it to handle a large volume of transactions in a single L1 commitment.

$$\text{HeaderHash} := \mathcal{H}_{\text{blake2b-224}}(\text{Header})$$

$$\text{Header} := \left\{ \begin{array}{ll} \text{prev.utxos_root} : & \text{RootHash} \\ \text{utxos_root} : & \text{RootHash} \\ \text{transactions_root} : & \text{RootHash} \\ \text{deposits_root} : & \text{RootHash} \\ \text{withdrawals_root} : & \text{RootHash} \\ \text{start_time} : & \text{PosixTime} \\ \text{end_time} : & \text{PosixTime} \\ \text{prev_header_hash} : & \text{HeaderHash} \\ \text{operator_vkey} : & \text{VerificationKey} \\ \text{protocol_version} : & \text{Int} \end{array} \right\}$$

These commitments are submitted by block producers who are selected via rotating consensus using an onchain linked list, who are each given a time window in which to submit however many commitments they like. This allows the system to record blocks at a rate independent of the L1's speed, and allows its participants to anticipate the behaviour of the L2. Combined with the deterministic fraud proving protocol, the practical speed with which Sundial can process transactions is only rate limited by the production of the operators and the processing of the offchain observer.

The deterministic nature of Cardano transactions allows Sundial fraud proofs to pinpoint the specific site of a transaction that violated Sundial's ledger rules, without having to look at any other parts of that transaction, any other unrelated transactions within the block, or any other blocks. This keeps fraud proofs and their onchain validation procedures small and efficient, which reduces the time and cost needed to submit fraud proofs when invalid blocks are detected, which makes it feasible for a wider group of people to police Sundial's blocks. In this way, Sundial significantly reduces fraud proof size relative to optimistic rollups used in Ethereum and other account-based blockchain ecosystems, where a much larger part of the global blockchain state needs to be inspected when constructing and verifying a fraud proof.

Fraud Proofs and Block Integrity

A central security pillar of Sundial Protocol is its fraud-proof mechanism, a core feature of its optimistic rollup architecture adapted to the unique advantages of the UTxO model. Fraud proofs enable any participant in the system to challenge an invalid state transition committed by a block operator. Unlike EVM-based rollups, where global state must be reconstructed to verify fraud, Sundial benefits from the deterministic and modular design of UTxO chains, allowing for targeted, efficient dispute resolution.

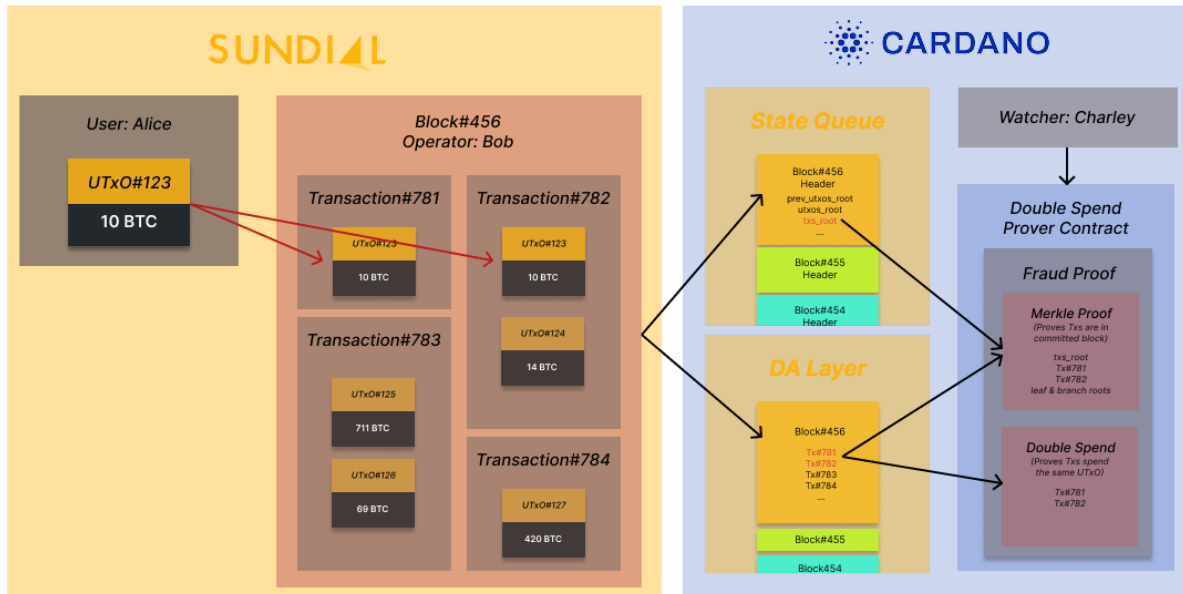
Operators proposing new rollup blocks must stake a significant bond to guarantee the validity of the block's transactions and resulting state root. Once posted, each block enters a defined challenge window during which any observer—known as a "watcher"—may submit a fraud proof. These fraud proofs must isolate the specific invalid transaction or state transition, citing protocol rule violations such as double spending, invalid signature verification, unauthorized UTxO creation, or incorrect fee deductions. Only the disputed transaction and its surrounding state context are recomputed on-chain, leveraging the stateless execution properties of UTxO systems to minimize validation costs.

To protect against denial-of-service or griefing attacks where malicious challengers attempt to slow down finality by submitting fraudulent disputes against honest blocks, Sundial requires both the operator and the challenger to bond collateral. If a fraud proof is valid, the operator's bond is slashed and partially awarded to the challenger. Conversely, if the proof is invalid, the challenger forfeits their bond. This bilateral bonding mechanism deters bad-faith reporting and ensures that only well-founded claims are submitted.

Consider a typical invalid block scenario:

Operator includes TX that spends UTx0#123 twice:

- User A's 10 BTC output is reused in two conflicting transactions.*
- Block root reflects only first spend; watcher detects absence of second output.*
- Fraud proof references TX index and UTx0 path; on-chain check confirms duplication.*



In the above example a dishonest operator constructs a block that includes two transactions, both attempting to spend the same UTxO output. Since the UTxO model enforces strict one-time use per output, the second transaction violates consensus rules. A watcher, observing this duplication, generates a fraud proof by submitting the transaction index, relevant Merkle proof path, and a minimal execution context to the L1 contract.

The hash of the first in a series of onchain scripts designed for validating this type of fraud is selected from an extensible fraud proof catalogue, and the verification begins at this step. These scripts form a computation thread (essentially a linear state graph) which passes from one script to the next as each one handles a granular piece of validation. At the final step the block is rolled back, and the operator penalized. This design allows for the ledger rules to be easily updated & enforced, and maximizes the simplicity of fraud proving by breaking it into minimal portions.

To ensure timely finality and efficient throughput, Sundial incorporates design insights from partners such as Anastasia Labs (L2 functionality), CheckPoint (security and chain optimization), and FluidTokens (L2 functionality and BTC integration inputs). Torram's planned contribution and collaboration with Sundial's infrastructure further supports real-time monitoring and light-client-compatible validation mechanisms, enabling decentralized fraud detection without relying on centralized oracles or heavy computational re-execution.

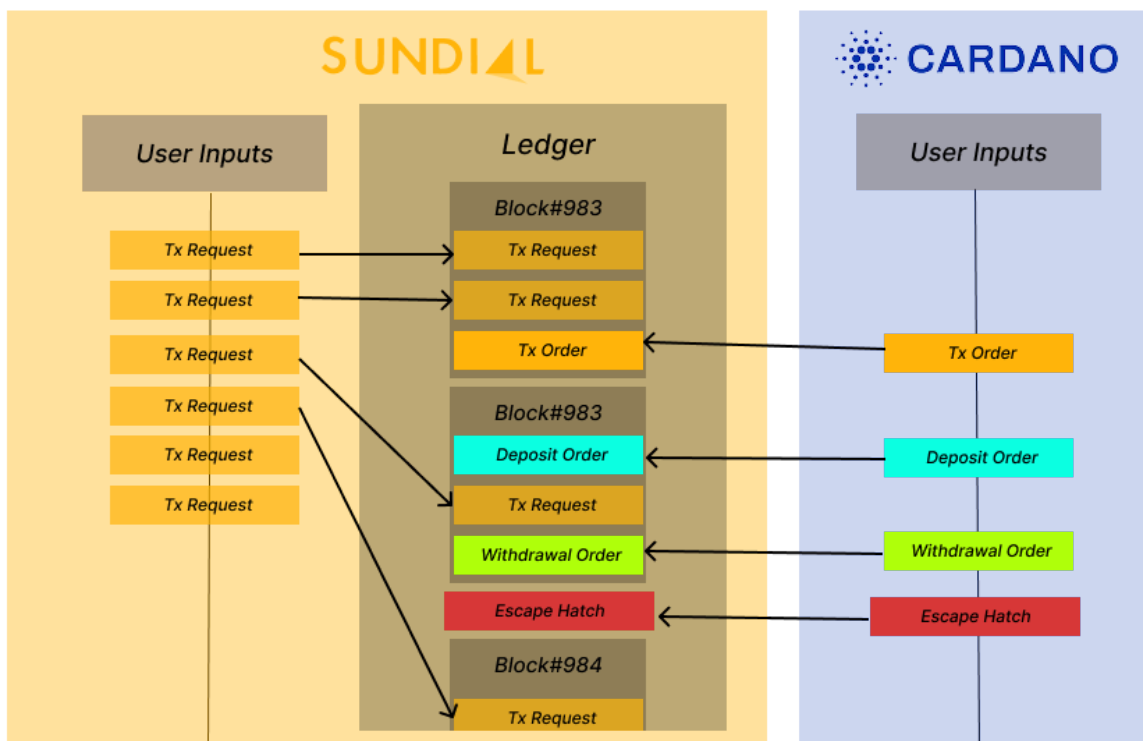
Ultimately Sundial is defined in two stages: off-chain inclusion (typically sub-minute) and on-chain finalization after the fraud challenge period expires. Because fraud proving is fully deterministic, apps can treat off-chain inclusion with the same degree of certainty as on-chain finality by simply running verification themselves. This layered approach ensures that the majority of honest blocks are confirmed quickly, while still enabling sufficient time for fraud detection and mitigation. To prevent the protocol from being exploited by mass challenges, future upgrades may adopt dynamic challenge windows or randomized verifier sampling.

Sundial's fraud-proof architecture reflects a philosophy of rigorous security, user-level transparency, and economic accountability. By tightly aligning incentives, leveraging the composability of UTxO transactions, and ensuring deterministic, stateless execution, Sundial protects its rollup integrity without compromising speed or user autonomy.

Censorship resistance and fallback mechanisms

On its own, the optimistic rollup mechanism described above ensures a high-level of assurance for the validity of block headers committed to the state queue and merged to Sundial's confirmed state. However, it does not prevent operators from censoring users' deposits, withdrawals, and L2 transactions. Consequently, Sundial's consensus protocol includes additional smart contract mechanisms to provide censorship resistance for these events.

Sundial deposits and withdrawals are initiated via L1 smart contracts that assign definite inclusion times to them. An operator block is invalid if it contains these inclusion times in its event interval but fails to include the associated deposit or withdrawal events. This ensures that if operators continue committing blocks to Sundial's state queue, then they cannot ignore deposit and withdrawal events. Sundial L2 transaction requests are typically submitted to operators via a publicly accessible API, and they can be ignored by operators. However, any user can escalate his L2 transaction request by posting a transaction order on L1. Similar to Sundial deposits and withdrawals, an L1 transaction order is assigned an inclusion time that guarantees its inclusion in a subsequent valid block.



If Sundial operators stop committing blocks at all to the state queue, then the inclusion times on their own cannot guarantee that deposits, withdrawals, and L2 transactions will be processed in a timely manner. However, for this extreme case, Sundial's consensus protocol

includes the escape hatch mechanism, which allows a special non-optimistic block to be appended to the state queue by a non-operator. This block can include any deposits, withdrawals, and L2 transactions that are verified on L1 to comply with Sundial's ledger rules. This ensures that user funds cannot be stranded on Sundial even if its operators entirely stop committing blocks.

Sundial Use Cases

The UTxO model, used by Bitcoin, Dogecoin, Litecoin, and other major blockchains, lacks native smart contract capabilities. As a result, trillions of dollars in hard assets remain idle, unable to generate yield or utility. Previous attempts at Bitcoin Layer-2 solutions, primarily using Ethereum Virtual Machine (EVM) architectures, have faced security vulnerabilities, technical limitations, and adoption challenges.

Sundial's primary use case - beyond scaling Cardano - is to seamlessly merge Bitcoin's vast liquidity with Cardano's advanced eUTxO smart contracts, enabling next-generation decentralized finance (DeFi). As the first sophisticated Layer-2 on Cardano, Sundial is purpose-built for scalability, reduced transaction costs, and institutional-grade security.

Key Innovations

Sundial introduces groundbreaking features to unlock new capabilities for UTxO-based blockchains:

- Babel Fees (Gas Abstraction) – Enables users to pay transaction fees with any token, enhancing DeFi accessibility and usability.
- ZK Bridges – A fully trustless rollup bridge secured by zero-knowledge proofs, ensuring seamless interoperability with metaprotocols.
- Native UTxO Security – Greatly diminish vulnerabilities common in other blockchain ecosystems. Such vulnerabilities include but are not limited to:
 - wallet drainers
 - smart contract exploits
 - fees for failed transactions
 - network outages
- Established Ecosystem Integrations – Day-1 ecosystem activity with leading DeFi, utility, RAW and gaming protocols coming from Cardano to drive adoption and usage.

Core Benefits

By integrating with Cardano Layer-1 and other UTxO chains, Sundial becomes a hybrid Layer-2 solution, delivering:

- Trustless UTxO Interoperability – Seamlessly connects Bitcoin, Cardano, and other UTxO-based assets.
- Trading – Enables low-cost, high-speed on-chain and cross-chain asset exchanges.
- Lending & Borrowing – Allows BTC, ADA, LTC, and other assets to be used as collateral in DeFi.

- Staking & Yield Generation – Implements secure and sustainable on-chain rewards mechanisms.
- DeFi & Web3 Integration – Positions BTC as a foundational asset for decentralized applications in finance, culture, and entertainment, leveraging Cardano's existing ecosystem.
- Institutional-Grade Compliance – Implements advanced compliance, reporting, and risk management tools to support institutional participation.

Sundial redefines the potential of UTxO assets, bridging the gap between Bitcoin and smart contract-driven DeFi, unlocking liquidity, and expanding the possibilities of blockchain finance.

Conclusion

Sundial is the first optimistic rollup on Cardano, designed to scale transactions while preserving Layer 1 security and decentralization. By serving as a hybrid Layer-2 for Bitcoin and other UTxO chains, Sundial creates a “super” UTxO network, unlocking novel use cases and seamless interoperability. With Bitcoin’s market capitalization projected to reach \$10 trillion by 2030, the need for scalable DeFi solutions is greater than ever. While the EVM ecosystem thrives on cross-chain connectivity, UTxO networks - despite being eight times larger - remain isolated. Sundial bridges this gap, enabling trustless asset movement, institutional-grade security, and the full integration of Bitcoin into decentralized finance. Positioned as a critical infrastructure layer in the evolving financial landscape, Sundial is set to redefine blockchain scalability and utility.